FEATURE ENGINEERING, MODEL OPTIMIZATION & PERFORMANCE COMPARISON

Artificial Intelligence & Machine Learning Internship - Task 2

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1. Introduction

The objective of this project is to improve house price prediction using machine learning techniques. The California Housing Dataset was selected because it contains multiple housing-related features such as income, house age, population, and geographical information. The project focuses on applying feature engineering, feature scaling, and model comparison techniques to determine which regression algorithm performs best. This task demonstrates a complete machine learning workflow including data preprocessing, model training, performance evaluation, and result interpretation.

2. Methodology

The project was completed using a structured machine learning pipeline. First, the California Housing Dataset was loaded and explored to understand the available features. Exploratory Data Analysis (EDA) was performed to examine feature relationships and data distribution. A new feature called RoomsPerPerson was created using Feature Engineering techniques. This feature was designed to represent room availability relative to population. After preprocessing, StandardScaler was applied to normalize the numerical values and improve model performance. The dataset was then divided into training and testing sets. Three machine learning models were trained and compared: Linear Regression, Ridge Regression, and Decision Tree Regressor.

3. Models Used

Linear Regression: Used as a baseline model to understand the relationship between housing features and house prices.

Ridge Regression: A regularized version of Linear Regression that helps reduce overfitting and improve generalization.

Decision Tree Regressor: A tree-based algorithm capable of capturing non-linear patterns and complex feature interactions.

4. Results and Performance Comparison

The performance of each model was evaluated using RMSE (Root Mean Squared Error) and R^2 Score. RMSE measures the prediction error, while R^2 Score measures how much variance in the target variable is explained by the model. Lower RMSE values and higher R^2 Scores indicate better performance.

Model	RMSE	R^2 Score
Linear Regression	0.745	0.578
Ridge Regression	0.745	0.576
Decision Tree Regressor	0.713	0.613

The Decision Tree Regressor achieved the best performance among all evaluated models. It produced the lowest RMSE value of 0.713 and the highest R^2 Score of 0.613. This indicates that the Decision Tree model was able to capture complex relationships in the dataset more effectively than the linear models.

5. Visual Analysis

Several visualizations were created during the project, including a Correlation Heatmap, Pairplot Analysis, Model Comparison Graphs, Actual vs Predicted Plot, Feature Importance Graph, and Residual Plot. These visualizations helped identify feature relationships, understand model behavior, and evaluate prediction quality.

6. Conclusion

This project successfully demonstrated the importance of Feature Engineering and Model Optimization in machine learning. By creating new features, applying feature scaling, and comparing multiple regression algorithms, the predictive performance of the model was analyzed in detail. The Decision Tree Regressor emerged as the best-performing model with the highest R^2 Score and lowest RMSE. The project provided practical experience in data preprocessing, feature engineering, machine learning model training, evaluation metrics, visualization, and performance comparison. These skills are essential for building reliable and efficient machine learning solutions.